Roll No. .....

Total Printed Pages: 03

CSA-102

Examination –May- 2024

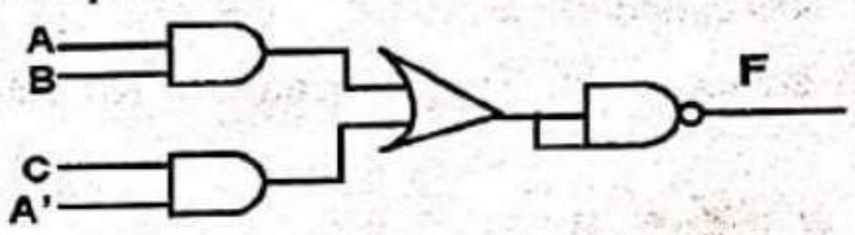
B.Tech. I Sem : CSE, IT, AIADS, CSE(BC), IoT

Digital Electronics

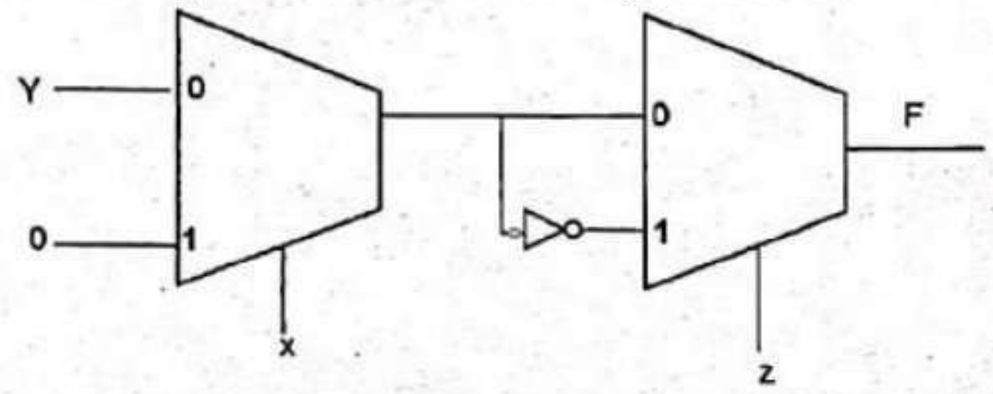
Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

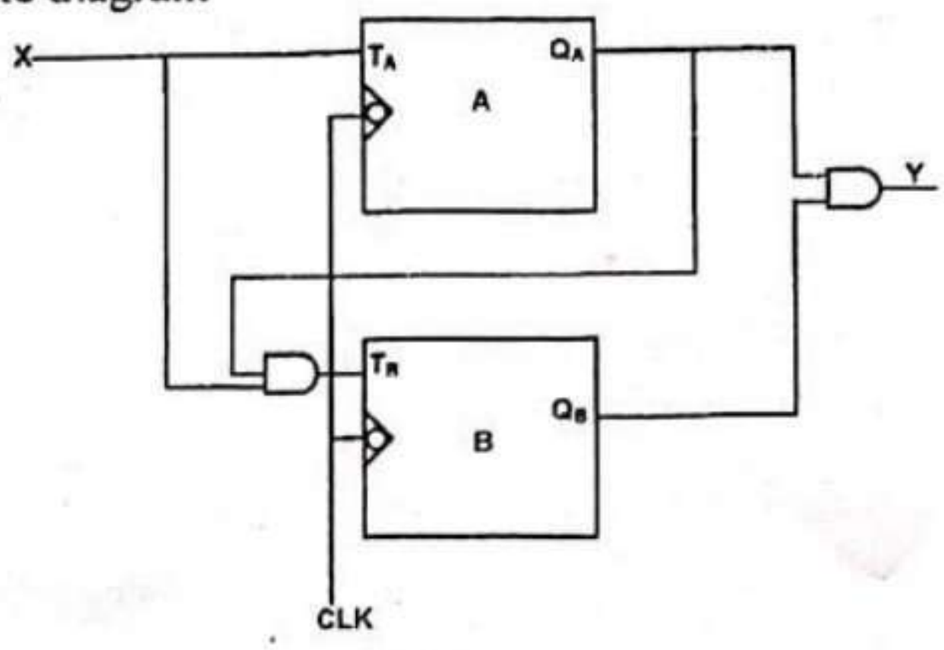
			Marks	BL	CO
Q.1	(a)	(i) How many bits are needed to represent a decimal number in the range of 0 to 63? (ii) What type of error can even parity detect? a) Single bit error b) Double bit error c) Three bit error d) All of the above	02	1,2	CO1
	(b)	Subtract (-15) with (-7).	02	2,3	CO1
	(c)	BCD Addition = $(475)_{BCD} + (198)_{BCD}$	03	2,3	CO1
		OR			
		Name the different types of parity. Explain any one.	03	2,3	CO1
	(d)	Convert the following : (a) $(43.23)_{10}$ to $(N)_8$ (b) $(ABC.67)_{16} + (DD5.89)_{16}$ (c) (10110111)2, the 2's complement of the no. is ..... (d) $(5371)_8$ to $(N)_{16}$ (e) $(110110)_2$ to gray code	05	4	CO1
		OR			
		Check for errors and correct if possible in the Hamming code: 1111011. Considered odd parity.	05	4	CO1
Q.2	(a)	The output F is  	02	2,3	CO2
	(b)	What is the simplified form of the Boolean expression: A'B + AB' + A'B'	02	2	CO2
	(c)	Explain the difference between NAND and NOR gates, and how they can be used to implement other logic functions.	03	3,4	CO2
		OR			
		Find SOP $F = \pi (0, 2, 5,8,12), d = (1, 10, 15)$	03	3,4	CO2

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	(d)	Design a combinational circuit that converts 3 bit binary no. into gray code.	05	5	CO2
		OR				
		Design a combinational circuit that multiply two binary numbers, each two bit long (a_o, a_1 & b_o, b_1) in order to form their product in binary.	05	5	CO2
Q.3	(a)	(i) An encoder has ……input and ……..output lines. (ii) In 8X1 multiplexer has ……..no. of select lines.	02	2,3	CO3
	(b)	Design the function using decoder $F1 = \Sigma (1,7)$ $F2 = \Sigma (0, 4, 3)$	02	3	CO3
	(c)	Explain the multiplexer with is diagram and truth table.	03	3,4	CO3
		OR				
		Two half adder and one not gate can implement a full adder, how?	03	3,4	CO3
	(d)	Two multiplexer are in cascaded, find the F.  	05	5	CO3
		OR				
		Implement a full substractor using decoder.	05	5	CO3
Q.4	(a)	(i) The output of a flip-flop can change state based on the presence of a/an: (a) Analog signal (b) Digital clock pulse (c) Continuously varying voltage (d) Mechanical switch (ii) A JK flip-flop toggles its output (Q) on every clock pulse when both J and K inputs are HIGH. This is called a: (a) Set condition (b) Reset condition (c) Toggle condition (d) Race-around condition	02	2,3	CO4
	(b)	Draw the SR flip Flop with NOR gate only.	02	2,3	CO4
	(c)	Convert D Flip Flop into J K Flip Flop.	03	3,4	CO4
		OR				
		Explain the working of T Flip Flop, with its diagram, truth table, excitation table and characteristic equation.	03	3,4	CO4

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(d) Draw the state diagram



05	5	CO4	

OR

Explain different type of shift register explain them with neat diagram. 05 | 5 | CO4

Q.5 (a) (i) A decade counter can count from _____ to _____. 02 | 2,3 | CO5

(ii) Which counter has the fastest operation?

a) Asynchronous counter b) Synchronous counter

c) Ripple counter d) Decade counter

(b) (i) In a decade counter, after reaching the state "1001" , what will be the next output? 02 | 2,3 | CO5

(ii) What is a synchronous preset counter, and what advantage does it offer?

a) It allows setting the initial count value before starting.

b) It simplifies the clocking scheme of the counter.

c) It increases the maximum counting speed.

d) It reduces the power consumption of the counter.

(c) Design a 3bit asynchronous down counter. 03 | 5 | CO5

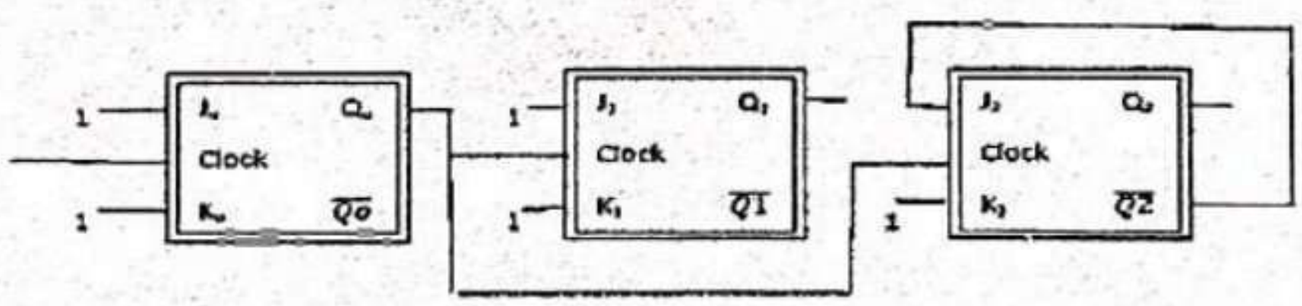
OR

Design a synchronous mode 5 up counter. 03 | 5 | CO5

(d) Design a synchronous counter that count 4, 1, 0, 7,6. 05 | 5 | CO5

OR

Find the next four state of the counter. If the initial state is 001 05 | 5 | CO5



Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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